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Ex No – 10A	Best Fit
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Program :

```
#include<stdio.h>

#include<string.h>

void bestFit(int blockSize[], int m, int processSize[], int n) {

    int allocation[n];

    memset(allocation, -1, sizeof(allocation));

    for (int i = 0; i < n; i++) {

        int bestIdx = -1;

        for (int j = 0; j < m; j++) {

            if (blockSize[j] >= processSize[i]) {

                if (bestIdx == -1) {

                    bestIdx = j;

                } else if (blockSize[bestIdx] > blockSize[j]) {

                    bestIdx = j;

                }

            }

        }

        if (bestIdx != -1) {

            allocation[i] = bestIdx;

            blockSize[bestIdx] -= processSize[i];

        }

    }

    printf("\nProcess No.\tProcess Size\tBlock no.\n");
```

```

for (int i = 0; i < n; i++) {
    printf("%d\t\t%d", i + 1, processSize[i]);
    if (allocation[i] != -1) {
        printf("\t\t%d\n", allocation[i] + 1);
    } else {
        printf("\t\tNot Allocated\n");
    }
}
}

int main() {
    int blockSize[] = {100, 500, 200, 300, 600};
    int processSize[] = {212, 417, 112, 426};
    int m = sizeof(blockSize) / sizeof(blockSize[0]);
    int n = sizeof(processSize) / sizeof(processSize[0]);
    bestFit(blockSize, m, processSize, n);
    return 0;
}

```

Output :

```

Process No.      Process Size      Block no.
1                212              4
2                417              2
3                112              3
4                426              5
[cse32@localhost ~]$

```

Program :

```
#include <stdio.h>

#define max 25

int main()
{
    int frag[max], b[max], f[max], i, j, nb, nf, temp;
    static int bf[max], ff[max];
    printf("\nEnter the number of blocks: ");
    scanf("%d", &nb);
    printf("Enter the number of files: ");
    scanf("%d", &nf);
    printf("\nEnter the size of the blocks:-\n");
    for (i = 1; i <= nb; i++)
    {
        printf("Block %d: ", i);
        scanf("%d", &b[i]);
    }
    printf("Enter the size of the files:-\n");
    for (i = 1; i <= nf; i++)
    {
        printf("File %d: ", i);
        scanf("%d", &f[i]);
    }
    for (i = 1; i <= nf; i++)
    {
```

```

    for (j = 1; j <= nb; j++)
    {
        if (bf[j] != 1)
        {
            temp = b[j] - f[i];
            if (temp >= 0)
            {
                ff[i] = j;
                frag[i] = temp;
                bf[j] = 1;
                break;
            }
        }
    }

    printf("\nFile_no:\tFile_size:\tBlock_no:\tBlock_size:\tFragment
");

    for (i = 1; i <= nf; i++)
    {
        printf("\n%d\t\t%d\t\t%d\t\t%d\t\t%d",
            i, f[i], ff[i], b[ff[i]], frag[i]);
    }

    return 0;
}

```

Output :

```
Enter the number of blocks: 4
Enter the number of files: 3
```

```
Enter the size of the blocks:-
```

```
Block 1: 5
```

```
Block 2: 8
```

```
Block 3: 4
```

```
Block 4: 10
```

```
Enter the size of the files:-
```

```
File 1: 1
```

```
File 2: 4
```

```
File 3: 7
```

File_no:	File_size:	Block_no:	Block_size:	Fragment
1	1	1	5	4
2	4	2	8	4
3	7	4	10	3[cse32@localhost ~]\$